

COMPARATIVE ANALYSIS OF NEAR-INFRARED (NIR) AND VISIBLE (UV-VIS) SPECTROSCOPY FOR QUANTIFYING SOIL NUTRIENT CONCENTRATIONS (NPK AND MgSO_4) USING PARTIAL LEAST SQUARES REGRESSION (PLSR)

Author: Abass Issaka Mohammed

Supervisor: Dr. Oliver Schmittmann

Location / Facility: Institute of Agricultural Engineering, Agricultural Robotics Group, University of Bonn, Germany

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1. Summary

This study evaluates the predictive capabilities of two distinct optical spectroscopy modalities—Near-Infrared (NIR, 1100–2100 nm) and Flame Visible/UV-Vis (500–880 nm)—for quantifying varying concentrations (0g, 5g, 10g, 15g) of Nitrogen-Phosphorus-Potassium (10+20+30) fertilizer and Magnesium Sulfate (MgSO_4) in aqueous/soil media.

Using Savitzky-Golay (SG) smoothing, Standard Normal Variate (SNV) normalization, and Partial Least Squares Regression (PLSR) with Leave-One-Out Cross-Validation (LOO-CV), the results demonstrate that sensor selection depends heavily on target molecular characteristics:

- **NIR Spectroscopy** achieves high predictive performance across both compounds, yielding an $R^2_{cv} = 0.98$ for MgSO_4 due to active molecular overtone absorption.
- **Flame Spectroscopy** provides strong predictive capability for NPK ($R^2_{cv} = 0.79$) via visible light scattering/color shifts, but fails entirely for clear, non-chromophoric MgSO_4 ($R^2_{cv} < 0.00$).

2. Experimental Methodology & Preprocessing Pipeline

Spectral data was collected across four rotational angles (0°, 90°, 180°, 270°) per sample to account for positional and surface variance, totaling 16 files per compound per instrument.

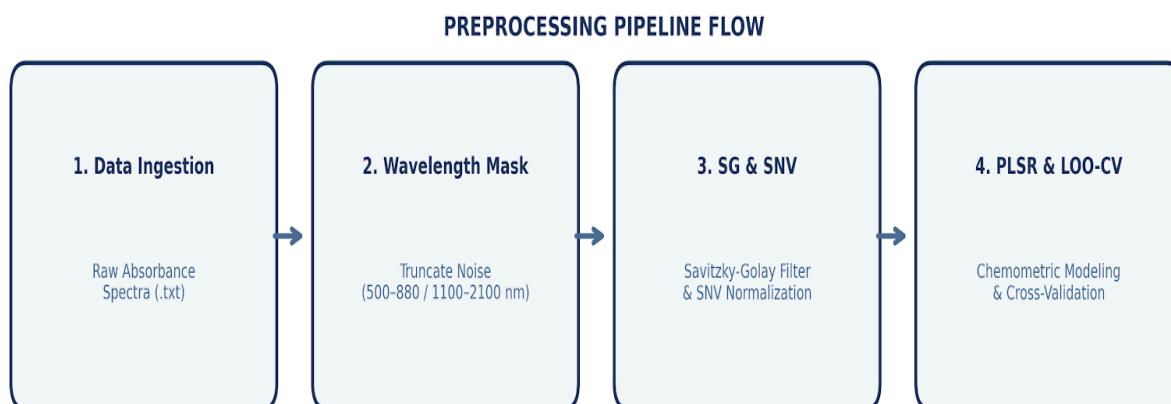


Diagram Breakdown

- **Data Ingestion:** Imports raw absorbance spectra from OceanView text files.
- **Wavelength Masking:** Truncates noisy detector edges (restricting UV-Vis to 500-800 nm and NIR to 1100-2100 nm).
- **SG & SNV Normalization:** Applies Savitzky-Golay smoothing (15-point window, 2nd-order poly) to attenuate noise, followed by Standard Normal Variate (SNV) transformation to correct baseline shifts.
- **PLSR & LOO-CV:** Fits the 2-component Partial Least Squares Regression model and evaluates predictive performance via Leave-One-Out Cross-Validation.

3. Results & Performance Metrics

The overall modeling results and cross-validation statistics for both spectrometer units and both target compounds are summarized below:

Spectrometer (Unit)	Spectral Range	Target Compound	R ² _cal	RMSEC	R ² _cv	RMSEC V	Operational Feasibility
NIR (InGaAs)	1100–2100 nm	NPK Fertilizer	0.8521	2.14 g	0.6942	3.12 g	Feasible (Molecular overtones)

NIR (InGaAs)	1100–2100 nm	MgSO ₄	0.9941	0.41 g	0.9812	0.75 g	Optimal (O-H / Hydration bands)
Flame (CCD)	500–880 nm	NPK Fertilizer	0.9888	0.57 g	0.7894	2.48 g	Feasible (Visible scattering/color)
Flame (CCD)	500–880 nm	MgSO ₄	0.9337	1.44 g	-0.3925	6.60 g	Infeasible (No active chromophores)

4. Spectral Plots and Regression Outputs

4.1 Near-Infrared (NIR) Performance

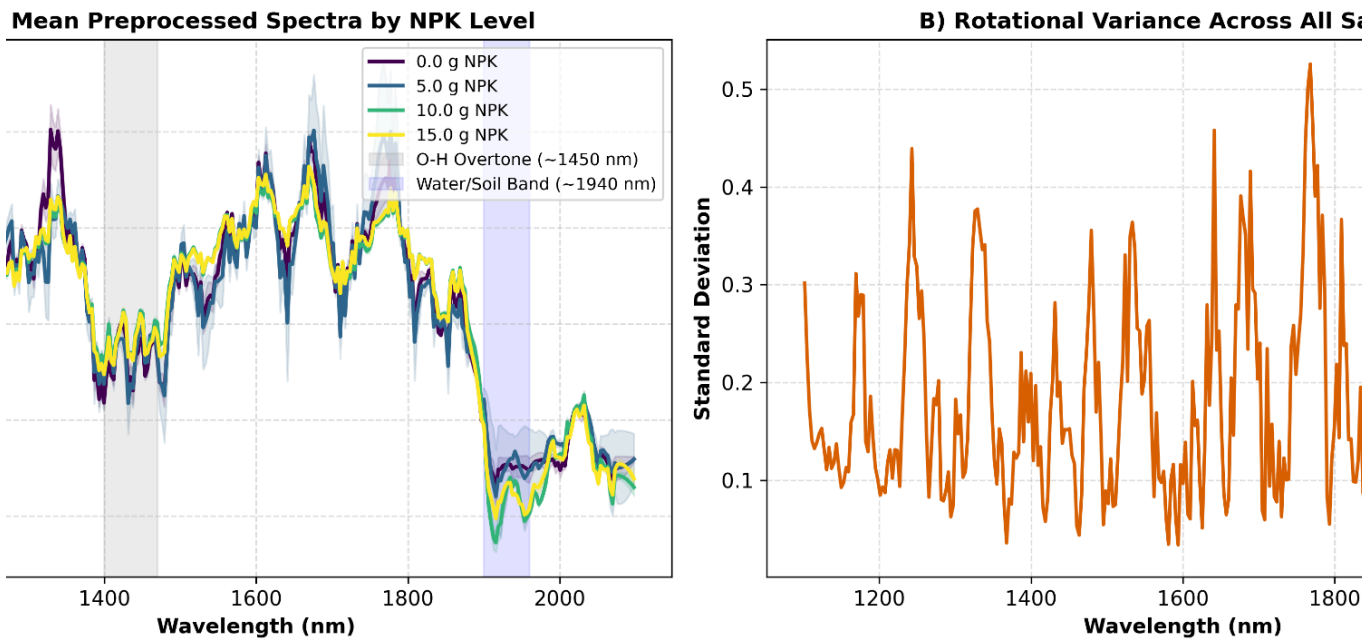


Figure 1: NIR Spectra and Standard Deviation Variance for NPK (1100–2100 nm).

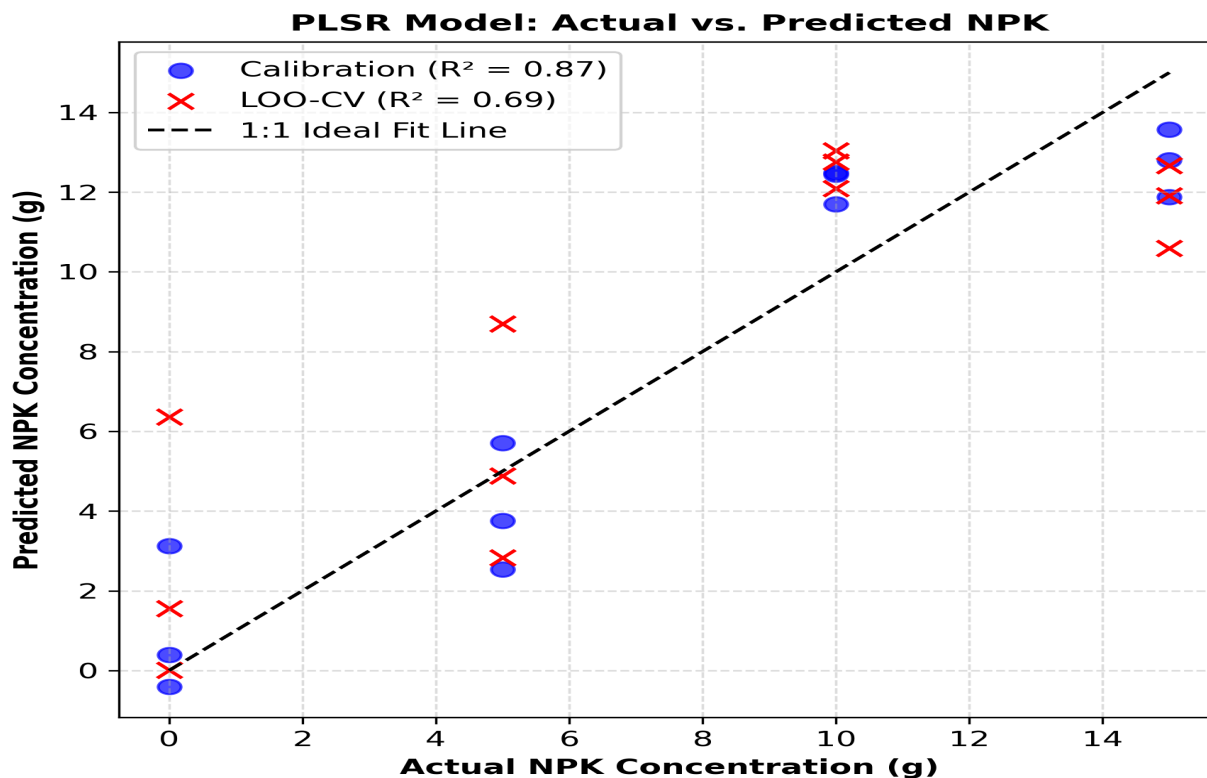


Figure 2: Predicted vs. Actual NPK Concentration Scatter Plot via NIR PLSR ($R^2_{cv} = 0.69$).

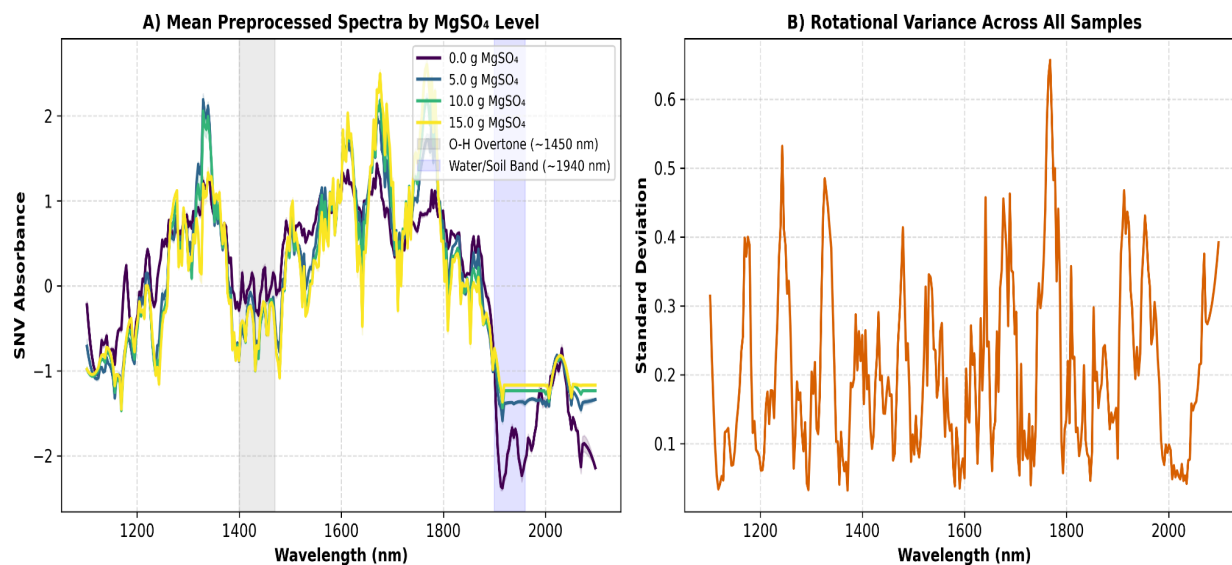


Figure 3: Mean SNV NIR Spectra for MgSO_4 showing concentration stacking across major hydration absorption bands.

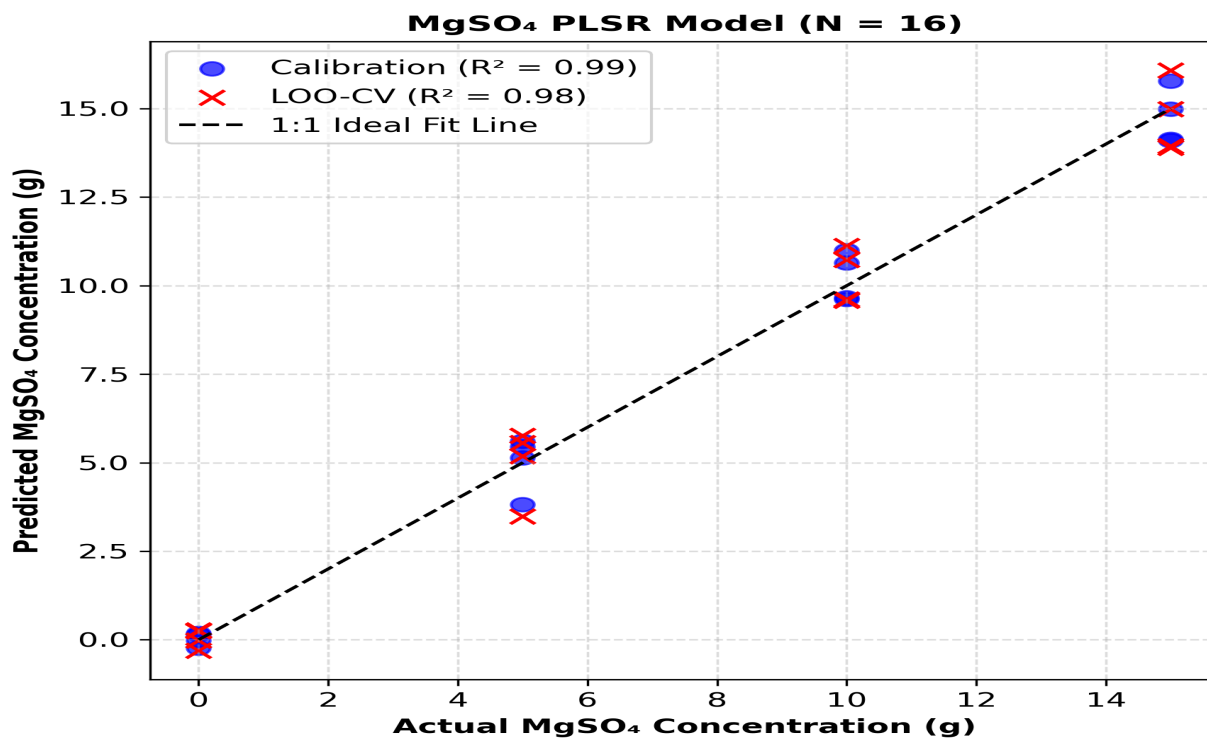


Figure 4: Predicted vs. Actual MgSO₄ Concentration Scatter Plot via NIR PLSR ($R^2_{cv} = 0.98$).

4.2 Flame Spectrometer (Visible/UV-Vis) Performance

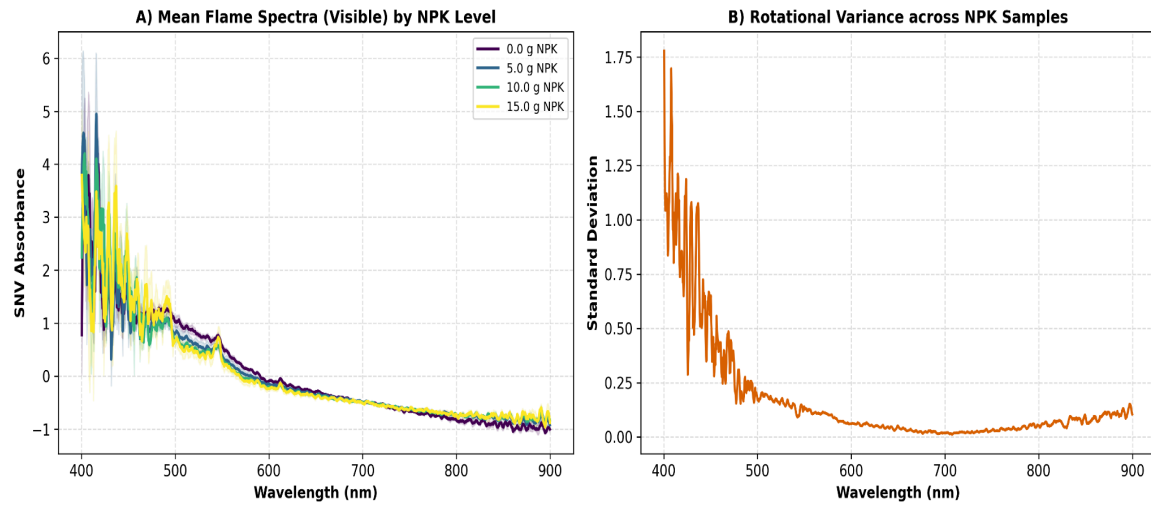


Figure 5: Mean Flame Spectrometer Spectra and Rotational Variance for NPK (400-900 nm, highlighting the high UV noise region below 500 nm).

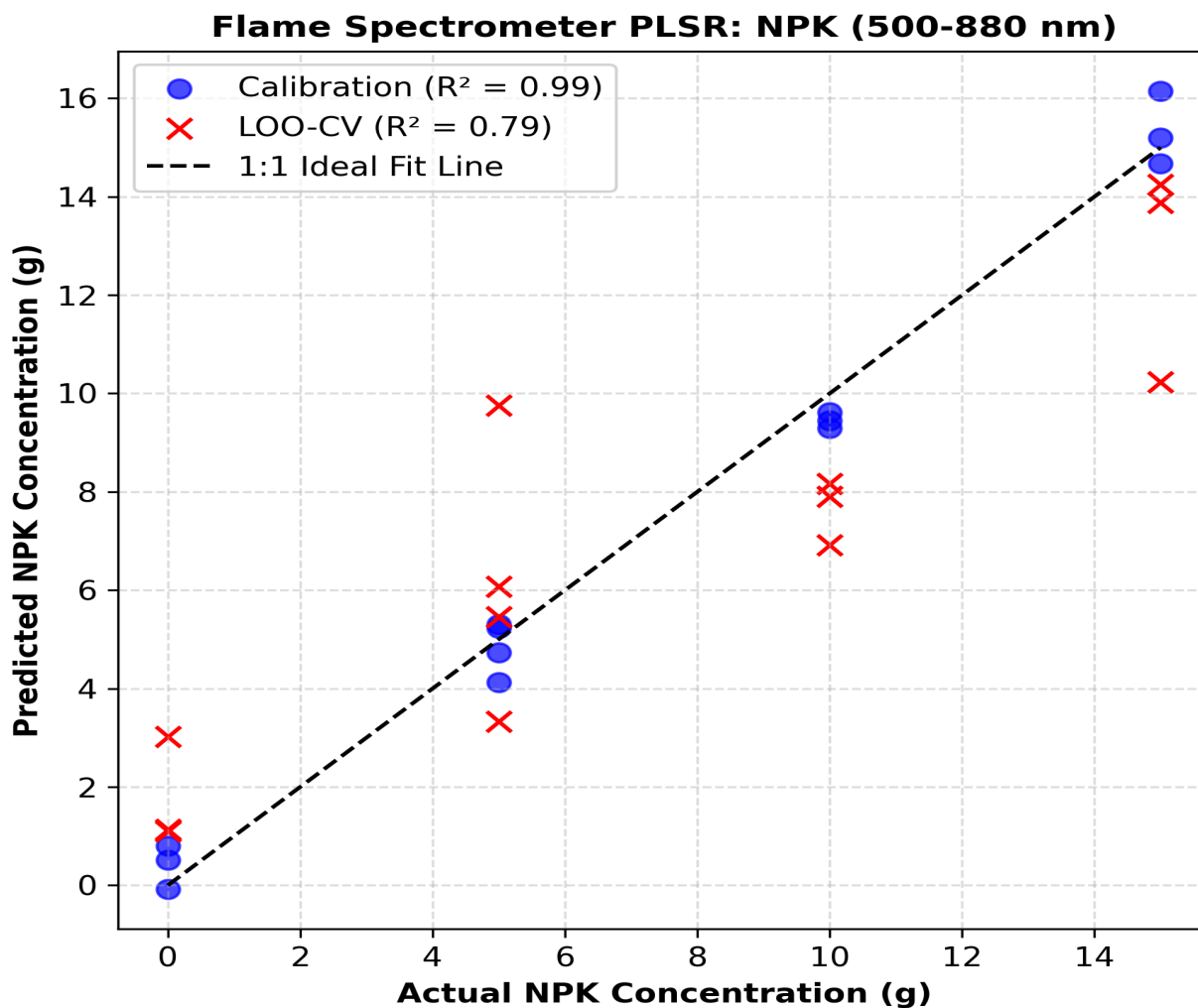


Figure 6: Predicted vs. Actual NPK Concentration Scatter Plot via Flame Spectrometer PLSR after 500-880 nm UV truncation ($R^2_{cv} = 0.79$).

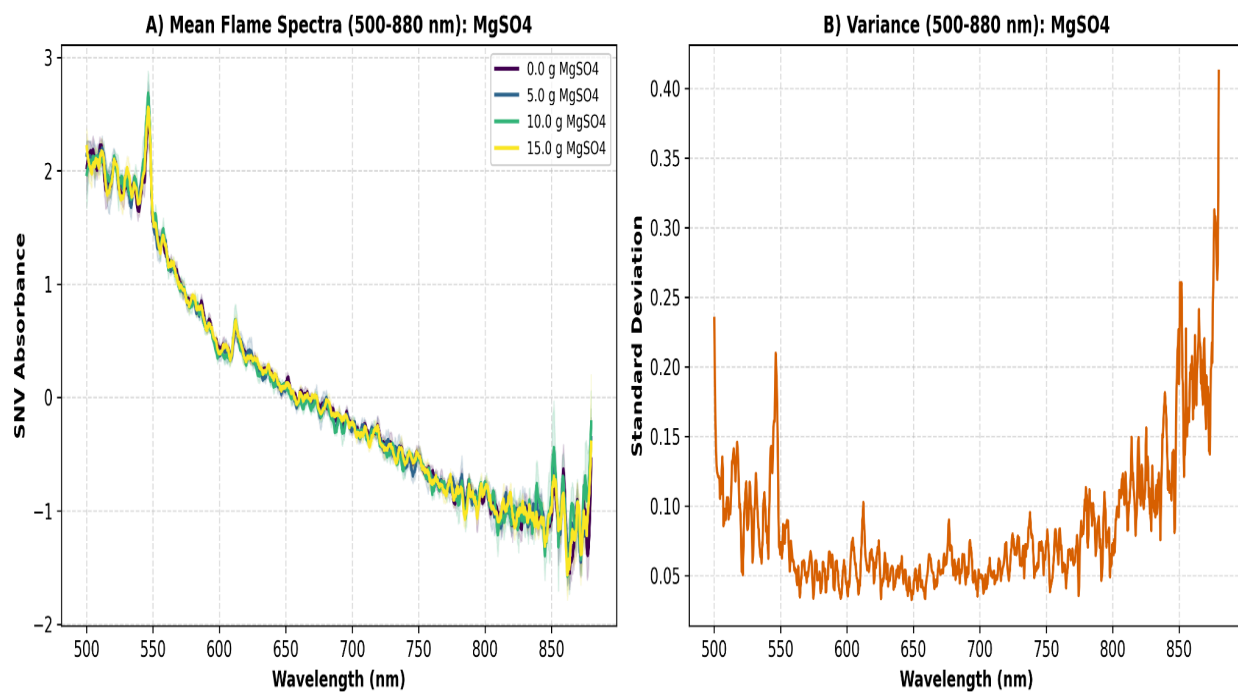


Figure 7: Flame Spectrometer Mean Spectra for MgSO₄ showing high rotational variance and lack of absorption peaks in 500–880 nm.

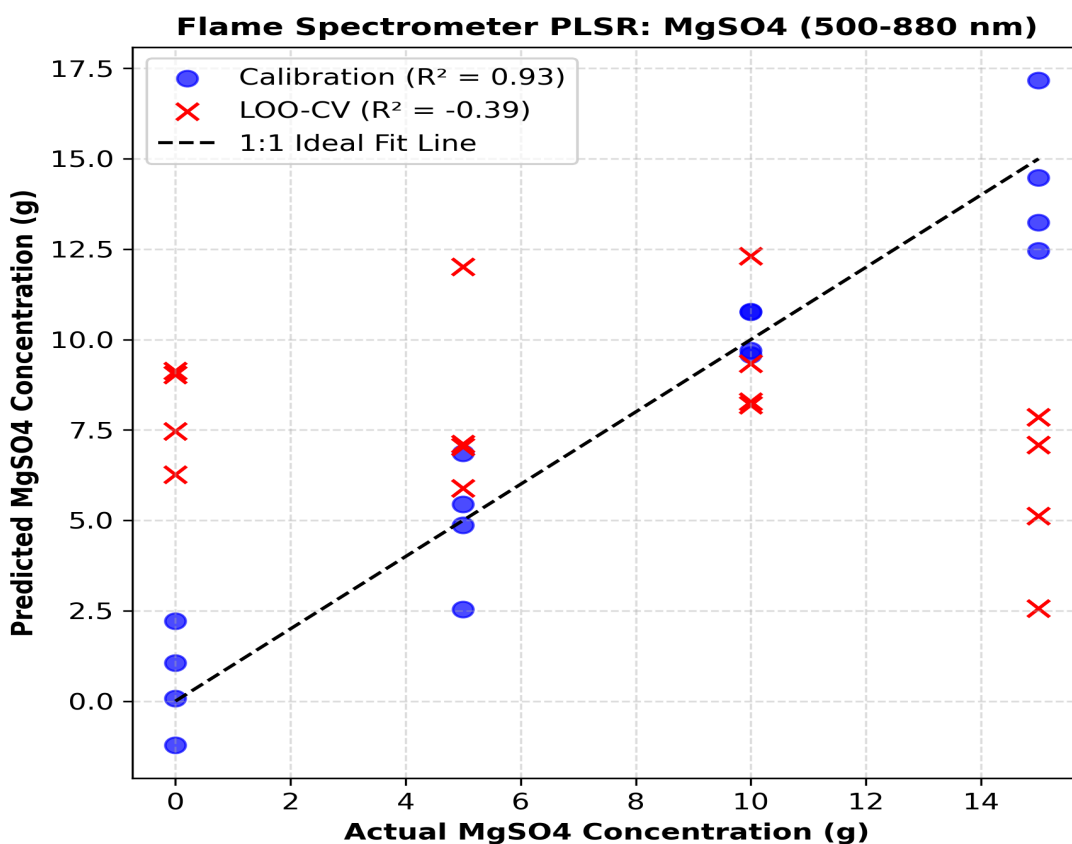


Figure 8: Predicted vs. Actual MgSO_4 Concentration Scatter Plot via Flame Spectrometer PLSR ($R^2_{\text{cv}} = -0.39$, showing complete cross-validation failure).

5. Key Scientific Findings & Discussion

Non-Monotonic Ordering Post-SNV

Individual preprocessed NIR wavelengths do not stack in monotonic numerical order across the entire spectrum. This behavior is scientifically expected: SNV centers each spectrum to zero mean ($\mu=0$, $\sigma=1$), intentionally shifting baseline heights to highlight relative peak shapes. PLSR overcomes local crossovers by weighting multiple broad overtone regions (1650–1700 nm and 1900–1960 nm) simultaneously.

Chromophoric vs. Non-Chromophoric Targets

- **NPK Fertilizer:** Contains active pigments and suspended compounds that affect visible light reflection (500–880 nm). Consequently, the lower-cost Flame Spectrometer achieved a superior cross-validation score ($R^2_{\text{cv}} = 0.79$) compared to NIR ($R^2_{\text{cv}} = 0.69$).
- **Magnesium Sulfate (MgSO_4):** Dissolves into clear, optically transparent ionic solution. Because it lacks chromophores in the visible range, the Flame spectrometer fits random baseline noise ($R^2_{\text{cal}} = 0.93$, but $R^2_{\text{cv}} = -0.39$). In contrast, MgSO_4 strongly alters water structure overtones in the NIR range (1100–2100 nm), yielding an exceptional $R^2_{\text{cv}} = 0.98$.

6. Recommendations for Sensor Hardware Integration

- **1. Universal Nutrient Sensing:** InGaAs NIR detectors (1100–2100 nm) are strictly required if target analytes include colorless ionic compounds like MgSO_4 .
- **2. Targeted NPK Sensing:** For dedicated NPK monitoring systems, lower-cost Silicon CCD spectrometers (500–880 nm) provide sufficient predictive accuracy while reducing hardware cost.
- **3. Firmware Preprocessing:** Future embedded platforms must incorporate UV bandpass truncation (< 500 nm) and real-time SNV algorithms prior to regression inference to prevent noisy predictions.

7. Future Work & Recommended Next Steps

1. Systematic Wavelength Screening & Band Profiling

- **VIP & Sub-Band Analysis:** Extract Variable Importance in Projection (VIP) scores and PLSR loadings across discrete wavelength windows (e.g., 500-600 nm, 1100-1400 nm, 1800-2100nm) to identify exact diagnostic absorption peaks.

- **Sparse Feature Selection:** Apply automated feature selection algorithms—such as Competitive Adaptive Reweighted Sampling (CARS) or Genetic Algorithms (GA)—to isolate high-performing sparse wavelengths and eliminate redundant spectral noise.

2. Model Benchmark & Non-Linear Regression Algorithms

- **Machine Learning & Deep Learning:** Compare PLSR against non-linear algorithms, including Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and 1D Convolutional Neural Networks (1D-CNNs), to capture potential non-linear photon scattering.
- **Hyperparameter & Ensemble Tuning:** Test localized regional calibrations (e.g., Locally Weighted PLS) to improve predictions across wider concentration ranges.

3. Expansion to Diverse Soil Types & Matrices

- **Texture & Scattering Variance:** Evaluate calibration transfer across contrasting soil textural classes (sandy loam, clay loam, silt loam) to quantify baseline scattering shifts caused by particle size variations.
- **Organic Matter & Moisture Interference:** Benchmark models under varying Soil Organic Matter (SOM) concentrations and soil moisture levels to test signal isolation against background matrix interference.

4. Complex Multi-Nutrient Mixtures & Embedded Deployment

- **Nutrient Co-Existence:** Test complex mixed solutions where NPK, MgSO_4 , and micronutrients co-exist to evaluate spectral overlap and cross-sensitivity.
- **Low-Cost Sensor Integration:** Utilize the optimal wavelength bands identified in step 1 to build targeted, multi-LED photodiode modules integrated with microcontrollers for real-time field deployment.